

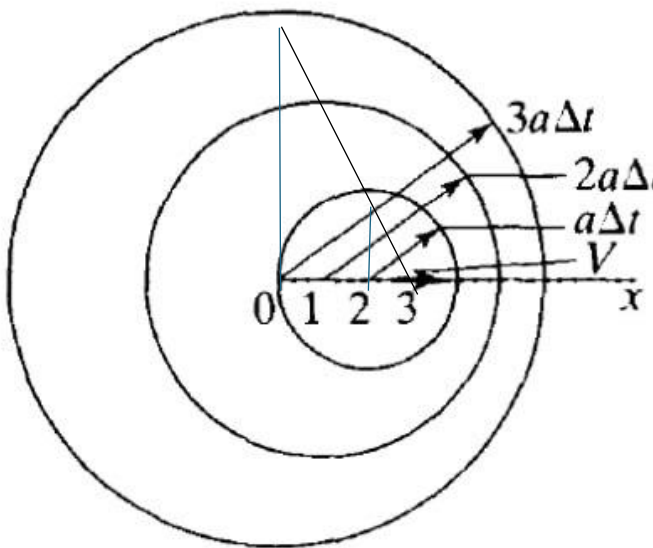
- 1- Air flows through a heat exchanger where its temperature increases from $T_1=300\text{ K}$ to $T_2=1200\text{ K}$. During the process, the pressure changes from $p_1=100\text{ kPa}$ to $p_2=150\text{ kPa}$. Assume that air behaves as an ideal gas with gas constant $R=0.287\text{ kJ/kgK}$, and that its specific heat at constant pressure c_p is temperature-dependent and given by:

$$c_p(T)=1.005+0.0005T \text{ (kJ/kgK)}$$

T is in Kelvin.

Calculate the change in specific entropy s_2-s_1 , use a numerical integration method to compute the integral term in entropy.

2. A supersonic airplane is flying at a constant altitude at a speed of $U=500\text{ m/s}$ at an altitude of $10,000\text{ m}$. The speed of sound at that altitude is $a=300\text{ m/s}$. A person is standing directly on the ground at a point where the airplane is flying overhead. Assuming the airplane flies in a straight line,
- How far horizontally (in meters) will the airplane have traveled before the person hears the sonic boom?
 - How long after the airplane passes overhead will the person hear the sonic boom?
3. Show that the flow speed in the picture below is $V=a/2$.



(d) $V = \frac{a}{2}$